# Dario Sanalitro
# Curriculum Vitae

Department of Electrical Electronic and Computer Engineering (DIEEI), Catania, Italy

dario.sanalitro@unict.it • https://dsanalit.github.io/
https://www.linkedin.com/in/dario-sanalitro-42a9b4151/

## Education

(From *October 2018* to *September 2021*) Ph.D. in *Robotics* at LAAS-CNRS and the Doctoral School Ècole Doctorale Systèmes (EDSYS) of the "Institut National des Sciences Appliquées de Toulouse" (INSA) in France. Degree awarded on April 8th, 2022. Thesis title: "*Aerial Cooperative Manipulation: full pose manipulation in air and in interaction with the environment*". Academic Advisor: Prof. Antonio Franchi.

(*May 16th 2019*) Test of English for International Communication (TOEIC).

(November 2017) Professional qualification as an Engineer.

(From *October 2015* to *October 2017*) Master's Degree in "*Automation Engineering and Control of Complex Systems*" from the University of Catania. Degree awarded on October 3rd, 2017, with a score of 110/110 cum laude. Thesis title: "*Digital particle image velocimetry analysis of RBCs flows in micro-channels*". Academic Advisor: Prof. Maide Bucolo.

(From *August 2015* to *November 2015*) Postgraduate Specialization Course "*Computer Expert in models and technologies for smart education*" organized by Links Management and Technology Spa as part of the EDOC@WORK3.0 Project, funded by the National Operational Programme for Research and Competitiveness 2007-2013 - Smart Cities and Communities and Social Innovation - Axis and Objective: Axis II - Integrated actions for sustainable development.

(From *October 2013* to *July 2015*) Bachelor's Degree in *Computer Engineering* from the University of Catania. Degree awarded on July 22nd, 2015, with a score of 109/110. Thesis title: "*Modeling and design of a bio-inspired system for underwater navigation*". Academic Advisor: Prof. Paolo Arena.

## Actual Position

(From *March 2023*) - Assistant Professor at the Department of Electrical, Electronics, and Computer Engineering (DIEEI) of the University of Catania under the project Sicilian Micro and Nano Technology Research and Innovation Center (SAMOTHRACE - E63C22000900006)

## Professional Experience

(From *November 2022 to February 2023*) Winner of a public selection competition for a research scholarship titled "*Brain Computer Interface Algorithms for the control of home automation systems*" issued by the Department of Electrical, Electronics, and Computer Engineering (DIEEI) at the University of Catania with Decree No. 2957 dated September 5th, 2022. Academic Advisor: Prof. Maide Bucolo.

(From *May 2022* to *October 2022*) Winner of a public selection competition for a Post-Doctoral scholarship issued by the LAAS-CNRS laboratory in Toulouse, France, with the research program "*Integration and experimental evaluation of an aerial robotic platform for object transportation and manipulation*". Academic Advisors: Simon Lacroix, Juan Cortés.

(From *October 2021* to *January 2022*) Fixed-term contract of employment at CNRS (French National Centre for Scientific Research) pursuant to Article 36 of Law No. 2020-734 dated June 17, 2020, carrying out research activities in continuation of the Ph.D. thesis.

(From *February 2022* to *April 2022*) Fixed-term contract of employment at CNRS (French National Centre for Scientific Research) carrying out research activities in continuation of the Ph.D. thesis.

(From *July 2018* to *September 2018*) Software Engineer at MODIS Consulting S.r.l. for Fiat Chrysler Automobiles (FCA). Location: Turin.

(From *December 2015* to *February 2016*) Software Developer at Links Management and Technology Spa. Location: Via Rocco Scotellaro, 55, 73100 Lecce LE, Italy.

## Participation in Funded Research Activities

The scholarship related to the Ph.D. activity was funded by the following project:

[PR-D1] (*2018-2022*) MuRoPhen Project funded by the French National Research Agency (ANR) (Project ANR-17-CE33-0007).

Additionally, during the Ph.D. course, collaborations and research activities were conducted in the following projects:

[PR-C1] (*2018-2022*) "Horizon 2020 Aerial Core" Project funded by the European Union (ID: 871479 AERIAL-CORE).

[PR-C2] (*2018-2022*) "Fly-crane: a multi-robot system for aerial transportation and manipulation" Project funded by the Occitanie region in 2018 under contract 2018 003431 - ESR PREMAT-000160.

## International Experiences

(From *May 2022* to *October 2022*) The Post-Doctoral activity was carried out at the LAAS-CNRS laboratories in Toulouse, Occitanie region, France, conducting research and experimental activities on the topic

"*Integration and experimental evaluation of an aerial robotic platform for object transportation and manipulation*" as part of the project. [PR-C2].

(From *October 2018* to *October 2021*) The Ph.D. course was carried out at the LAAS-CNRS laboratories in Toulouse, Occitanie region, France, conducting research on the topic "*Aerial Cooperative Manipulation*" within the scope of the projects MuRoPhen [PR-D1], Aerial Core [PR-C1], and Flycrane [PR-C2].

## Review Activities

Associate Editor for International Scientific Journals

**Nonlinear Engineering, Modeling and Application** since June 2023

Review Editor for International Scientific Journals

**Frontiers in Neurorobotics** since July 2023

Reviewer for international scientific journals

**IEEE Transactions on Control Systems Technology**

**IEEE Robotics Automation-Letters**

**IEEE Transactions on Robotics**

**IEEE Transactions on Systems, Man and Cybernetics: Systems**

**Elsevier Control Engineering Practice**

Reviewer for international scientific conferences

**IEEE International Conference on Robotics and Automation (ICRA)**

**IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**

**IEEE International Conference on Unmanned Aircraft Systems (ICUAS)**

**IEEE International Conference on Safety, Security, and Rescue Robotics (SSRR)**

## Participation in International Schools

(From *3rd to 7th July 2023*) Participation in the training offered by the Swartz Center for Computational Neuroscience (SCCN) University of California San Diego (UCSD) and CNR, titled "EEGLAB Workshop" with a duration of 32 hours.

(From *1st to 5th March 2021*) Participation in the training offered by CNRS titled "Language C++14/17" with a duration of 35 hours.

(*June 2019*) Participation in the training offered by the EDSYS Doctoral School titled "Pedagogical Practices for the University" with a total duration of 18 hours.

(From *2nd to 5th July 2019*) Participation in the training course titled "*Mathematical Theory of Grasping and Manipulation*" held by Prof. Domenico Prattichizzo from the University of Siena for a total duration of 20 hours.

(From *26th June* to *1st July 2019*) Participation in the "*ETH Robotics Summer School*", Zurich, Switzerland.

## International Competitions

[CO-1] (From *23rd to 27th February 2020*) Participation in the international robotics competition *MBZIRC*, Abu Dhabi, United Arab Emirates, with the team LAAS-CNRS.

## Speaker at International Conferences and Scientific Events

(From *29th May* to *2nd June 2023*) IEEE International Conference on Robotics and Automation (ICRA), London, UK. Presentation of the work: **D. Sanalitro**, M. Tognon, A. E. Jimenez-Cano, J. Cortés, and A. Franchi, "*Indirect Force Control of a Cable-Suspended Aerial Multi-Robot Manipulator*"

(From *31st May* to *4th June 2020*) IEEE International Conference on Robotics and Automation (ICRA), Paris, France. Virtual presentation of the work: **D. Sanalitro**, H. J. Savino, M. Tognon, J. Cortés, and A. Franchi, "*Full-pose manipulation control of a cable-suspended load with multiple UAVs under uncertainties*". https://www.youtube.com/watch?v=3fQV2JNzZF0

(From *1st* to *4th September 2020*) IEEE International Conference on Unmanned Aerial Vehicles (ICUAS), Athens, Greece. In-person presentation of the work: A. Petitti, **D. Sanalitro**, M. Tognon, A. Milella, J. Cortés, and A. Franchi, "*Inertial estimation and energy-efficient control of a cable-suspended load with a team of UAVs*"

(*May 16th 2022*) Presentation at the European Researchers' Night organized by Université Fédérale - Toulouse Midi-Pyrénées.

## Research Activity

(From *2017*) Since 2017, I have been involved in four main lines of research.

The first research line, which is currently in development at the University of Catania under the Sicilian Micro and Nano Technology Research and Innovation Center (SAMOTHRACE) project, deals with the miniaturization of biomedical devices for cellular analysis on micrometric scales.

The second one, which has been the focus of the PhD research activities and is still in development, aims at the development and application of methodologies and technologies for estimation and control of robotic systems, mainly aerial vehicles. The main research efforts have been carried out at the "Laboratoire d'analyse et d'architecture des systèmes" (LAAS) of CNRS in Toulouse, France.

The third one focuses on the experimental validation of bio-inspired methodologies applicable to robotic systems'.

These three main lines of research have also resulted from the initiation and development of collaborations, often characterized by the combination of complementary expertise, involving researchers from national and international Research Centers and Universities.

The research outcomes have been published in one monograph, six papers published in international scientific journals, and six papers published in the proceedings of international congresses.

### Aerial Cooperative Manipulation

In collaboration with:

(*2019*) Centro di Ricerca E. Piaggio, Dipartimento di Ingegneria dell'Informazione, Università degli Studi di Pisa. Head: Prof. Lucia Pallottino

(*2020*) Centro Sistemi e Tencologie Industriali Intelligenti per il Manifatturiero Avanzato, Consiglio Nazionale delle Ricerche (STIIMA CNR). Head: Prof. Antonio Petitti

(*2020*) Università degli Studi di Roma La Sapienza, Dipartimento di Ingegneria Informatica, Automatica e Gestionale (DIAG). Head: Prof. Giuseppe Oriolo.

(*2023*) University of Twente, Robotics and Mechatronics Lab. Head: Prof. Antonio Franchi.

One of the major capabilities of robotic systems, more specifically of aerial vehicles, is their ability to contribute to the perception of the surrounding environment through the ensemble of sensors with which they can be equipped. This capability has been widely exploited since their first appearance. The advent of Unmanned Aerial Vehicles (UAVs) has accelerated this process, as these systems can reach broader or inaccessible spaces compared to other categories of robots. Moreover, in recent years, technological progress has led to the use of robotic systems in increasingly complex operations, where robots have the capacity and potential to interact with the external and/or surrounding environment. This challenge has proven to be particularly relevant in many robotic applications such as grasping, transportation, positioning, assembly, and dismantling.
Although individual aerial vehicles could perform such tasks, and the wide variety of solutions proposed

in the state of the art demonstrates this, in some cases, the covered area and the overall payload of a single robot are limited. Therefore, the use of multi-robot systems allows overcoming such limitations. In this context studies have been conducted on the *cooperative manipulation problem* within the projects [PR-D1] and [PR-C1]. The main objective of these projects is to control one or more degrees of freedom of the load supported by the components of the multi-robot system.
In particular, models and estimation and control algorithms have been defined and experimentally validated to solve this problem in two specific contexts, encompassing a wide range of applications: one where there is no interaction with the surrounding environment and another where there is interaction with the surrounding environment.

As part of the first set, which involves models and algorithms without interaction with the surrounding environment, the research included the study of a dynamic formulation of the problem. The generality of this formulation lies in the type and number of robots involved and the types of connections that can be used. A decentralized control strategy based on a "leader-follower" paradigm and the analysis of its stability completed the study [IJ-6]. The experimental validation and the effect of model uncertainties and force biases in real-world scenarios has been studied in [IJ-1] and [IC-1].
Furthermore, within the scope of the aforementioned projects, a second kinematic formulation of a specific cable-driven configuration, called "Flycrane," was studied. This configuration utilizes the minimum set of robots to achieve full control of the supported load. The control strategy takes into account possible model uncertainties and dynamic disturbances, and its experimental validation was the subject of a second study [IJ-5].
In addition, an extension of this formulation for the online estimation of certain dynamic parameters, such as the mass and center of mass of the load, was the subject of a third study [IC-4].
The second set includes methodologies developed and experimentally validated to handle direct interactions with the surrounding environment, which can be represented by the environment itself, other robots, or human beings.
Among these methodologies, the first study focuses on a shared control strategy with a human operator for performing "pick and place" actions in the context of assembly and disassembly operations at construction sites [IJ-3]. This study is part of the [PR-C2] project.
The second study, on the other hand, concerns a control strategy that enables trajectory generation for a multi-robot system, considering the exchange of forces and moments between the system and its surroundings [IJ-4].
The study [IC-3] represents a contribution to the field of cooperative robotics, specifically in the context of coordination and planning for multi-robot systems executing "pick and place" actions. The research explores two distinct cooperative approaches, one based on communication among robots and the other communication-free. Through simulations, the proposed methodologies have been validated, demonstrating their effectiveness and potential applicability in real-world scenarios. The development of these cooperative strategies took place within the framework of a larger project aimed at participating in the prestigious international robotics competition, MBZIRC [CO-1].
The methodological, analytical, and experimental results obtained and documented in [M-1] demonstrate the significance and effectiveness of the developed models and control strategies to the extent that they can encourage their real-world application.
Nevertheless, there are several possible improvements that can be applied to multi-robot systems. One significant area that deserves attention is the decoupling of these systems from global positioning means, which are often present in indoor environments such as laboratories. Achieving this decoupling would pave the way for studying new approaches for estimating the relative positioning of all components

within multi-robot systems and developing communication algorithms among them.

## Modeling of Microfluidic Processes and Applications

In collaboration with:

(*2017*) Microfluidics Group - Università degli Studi di Catania. Head: Prof. Maide Bucolo

(*2017*) Microhemodynamics Lab - University of California San Diego (UCSD), USA. Head: Prof. Marcos Intaglietta

The manipulation of small quantities of fluids and micrometric particles suspended within microfluidic devices for the analysis of biological and chemical processes on portable and low-cost chips, such as "System-on-a-Chip," represents a relevant theme in the development of biotechnologies.
Of particular interest in this field are biphasic microfluidic processes, achieved by merging two immiscible fluids or suspended particles/cells (e.g., biological fluids) within a microchannel with diameters ranging from micrometers to millimeters.
These processes fall into the class of nonlinear systems. As a matter of fact, depending on the experimental conditions (physical, chemical, electrical, and optical properties of both phases and the device) and the designed microchannel geometry (straight, curved, with converging or diverging paths and chambers), the interaction between the two phases or the behavior of suspended particles/cells can vary significantly along the channel, leading to both linear and nonlinear behaviors. To ensure high precision and, above all, repeatability of these processes, it is essential to have technologies for flow management and monitoring that are low-cost, non-invasive, and easy to use. Additionally, flexible and accurate modeling methodologies are crucial for developing suitable and robust control schemes.
The activity developed in this context during my Master's degree program focused on the development of a methodology for monitoring particles, specifically red blood cells, in microchannels. This procedure is based on image analysis and involves studying the velocity of these particles under predetermined flows within the microchannels.
The developed methodology aimed to analyze hemodynamic processes at the microcirculatory level, and its validation was made possible through data obtained in collaboration with the Microhemodynamics Lab (UCSD).
Another project involved the study of a compact total-on-chip velocity detection approach for the label-free characterization of cell populations of different nature subjected to hydrodynamic stimuli [IC-2].
The future developments in this field involve different areas of investigation. Methodologically, there is a need for greater scientific awareness in understanding the impact of geometries, diameters, and materials of the microchannels, as well as the specific fluid properties influencing the transition from one behavior to another. This requires interdisciplinary work to gain deeper insights into the underlying phenomena and optimize the microfluidic systems for specific applications.
From a technological perspective, further efforts from the scientific community are necessary to engineer these devices with modular and cost-effective systems. The development of low-cost microfluidic devices will make them more accessible and widely applicable in various fields, such as medicine, biotechnology, environmental monitoring, and point-of-care diagnostics.

### Estimation of robot body parameters through bio-inspired models

In collaboration with:

(*2018*) Biorobotics Group - Università degli Studi di Catania. Head: Prof. Paolo Arena

Given the continuous evolution in the field of robotics, the knowledge and representation of the space occupied by a robotic system in relation to the surrounding environment are key elements in the development of high-level cognitive activities. The robot's spatial model can be represented through sensory and movement representations, and this representation can be used to perform various tasks. Examples include assessing the reachability or navigability of the surrounding environment through the analysis of its own movement, combined with perception through sensory organs.
In this context, in collaboration with the University of Catania, and given the promising results of methods based on learning techniques, neural networks, and bio-inspired neural networks in solving such problems, a study was conducted involving the development of a bio-inspired neural network known as a "spiking neural network." This network was applied to a humanoid robot (DarwinOp) for the estimation and learning of the relative distance between the robot and objects in the surrounding environment through the computation of parallax [IC-5].
The mentioned approach was validated through an experimental campaign that involved two main phases. The first phase consisted of training the bio-inspired neural network. During this training phase, the neural network was exposed to various inputs representing different distances between the humanoid robot's limbs and the surrounding objects. The network learned to compute parallax and estimate the relative distances between the robot and the objects based on these inputs. In the second phase, the trained neural network was applied in a real-world scenario.

Future developments indeed foresee the extension of neural networks in a way that learning about the occupied space relative to the surrounding environment can involve not only individual parts of the system but the entire robotic system as a whole.

### Brain-Computer Interfaces for Robotics Applications

In addition, I am currently pursuing a new research line focused on controlling robots using brain signals. The main idea behind this research is to establish a direct connection between the human brain and robotic systems through brain-computer interfaces (BCIs). In this direction a study has been published. In particular, it explores the integration of brain-computer interfaces (BCIs) with home automation systems [IJ-2].

## Scientific Publications

### Monographs

[M-1] **D. Sanalitro**, "Aerial cooperative manipulation: full pose manipulation in air and in interaction with the environment," Ph.D. dissertation, INSA de Toulouse, 2022.

### Publications on Internationals Journals (IJ)

[IJ-1] C. Gabellieri, M. Tognon, **D. Sanalitro**, A. Franchi, "Equilibria, Stability, and Sensitivity for the Aerial Suspended Beam Robotic System Subject to Parameter Uncertainty" accepted to IEEE Trans-

action on Robotics, 2023.

[IJ-2] S. Cariello, **D. Sanalitro**, A. Micali, A. Buscarino, M. Bucolo, "Brain–Computer-Interface-Based Smart-Home Interface by Leveraging Motor Imagery Signals", MDPI Inventions, vol. 8, no. 4, pp. 91, 2023.

[IJ-3] A. Jiménez-Cano, **D. Sanalitro**, M. Tognon, A. Franchi, and J. Cortés, "Precise cable-suspended pick-and-place with an aerial multi-robot system," Journal of Intelligent & Robotic Systems, vol. 105, no. 3, pp. 1–13, 2022.

[IJ-4] **D. Sanalitro**, M. Tognon, A. Jimenez-Cano, J. Cortés, and A. Franchi, "Indirect force control of a cable-suspended aerial multi-robot manipulator," IEEE Robotics and Automation Letters, 2022.

[IJ-5] **D. Sanalitro**, H. J. Savino, M. Tognon, J. Cortés, and A. Franchi, "Full-pose manipulation control of a cable-suspended load with multiple UAVs under uncertainties," IEEE Robotics and Automation Letters, vol. 5, no. 2, pp. 2185–2191, 2020.

[IJ-6] C. Gabellieri, M. Tognon, **D. Sanalitro**, L. Palottino, and A. Franchi, "A study on force-based collaboration in swarms," Springer, Swarm Intelligence, vol. 14, pp. 57–82, 2020.

## Publications in Proceedings of International Conferences (IC)

[IC-1] C. Gabellieri, M. Tognon, **D. Sanalitro**, A. Franchi, "Force-based Pose Regulation of a Cable-Suspended Load Using UAVs with Force Bias" accepted to 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

[IC-2] E. Cutuli, G. Stella, **D. Sanalitro**, L. Saitta, F. Guarino, G. Cicala, M. Bucolo, "3D-printed micro-optofluidic chamber for cell population characterization and velocity detection", Accepted to the 27th International Conference on Miniaturized Systems for Chemistry and Life Sciences, 2023 (microTAS)

[IC-3] E. Umili, M. Tognon, **D. Sanalitro**, G. Oriolo, and A. Franchi, "Communication-based and communication-less approaches for robust cooperative planning in construction with a team of UAVs," International Conference on Unmanned Aircraft Systems (ICUAS). IEEE, 2020, pp. 279–288.

[IC-4] A. Petitti, **D. Sanalitro**, M. Tognon, A. Milella, J. Cortés, and A. Franchi, "Inertial estimation and energy-efficient control of a cable-suspended load with a team of UAVs," International Conference on Unmanned Aircraft Systems (ICUAS). IEEE, 2020, pp. 158–165.

[IC-5] P. Arena, L. Patané, **D. Sanalitro**, and A. Vitanza, "Insect-inspired body size learning model on a humanoid robot," 7th IEEE International Conference on Biomedical Robotics and Biomechatronics (Biorob). IEEE, 2018, pp. 1127–1132.

[IC-6] F. Cairone, **D. Sanalitro**, M. Bucolo, D. Ortiz, P. J. Cabrales, and M. Intaglietta, "DPIV analysis of RBCs flows in serpentine micro-channel," European Conference on Circuit Theory and Design (ECCTD). IEEE, 2017, pp. 1–4.